

MBA in AI for Business — Análise Macro
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Parliamentary Expense Radar: Statistical Prioritization with LLM Triage and Human Validation

A reproducible CRISP-DM pipeline to prioritize — never accuse —
anomalous CEAP expenses

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ABSTRACT

The Parliamentary Activity Exercise Allowance — CEAP — totals 284,845 expense records and R\$345,090,001.79 over 19 months (Jan/2025 to Jul/2026), far beyond what any single person can review entry by entry. This work builds a complete CRISP-DM pipeline in three stages. First, it detects anomalous spending patterns with three independent, robust statistical methods (median/MAD, supplier-concentration HHI), reducing 601 legislators to 253 flagged by 2+ methods and 26 flagged by all three simultaneously. Second, it uses an LLM (Claude, via the Anthropic API) with Pydantic-structured output to classify severity and explain each case in citizen-facing language, without ever judging anyone's reputation. Third, it evaluates the LLM across multiple layers: consistency against a deterministic rubric and agreement against genuine human labeling (Cohen's kappa = 0.231; 40% accuracy). The evaluation's central finding is an LLM collapse bias: 70% of cases were classified as "medium" severity and 84% as "combination" category, almost independently of the actual signal — not random noise, but a systematic tendency toward "safe" answers. The average cost per classified case is US\$0.00525 (model claude-sonnet-4-6), making it cheap to scale classification to thousands of cases. Every number cited in this abstract is also generated by an executable code chunk in the body of this document.

Keywords: Parliamentary Allowance. Anomaly Detection. Language Models. Structured Output. CRISP-DM.

RESUMO

A Cota para o Exercício da Atividade Parlamentar — CEAP — soma 284.845 despesas e R\$ 345.090.001,79 em 19 meses (jan/2025 a jul/2026), longe do que qualquer pessoa consegue revisar despesa por despesa. Este trabalho constrói um pipeline CRISP-DM completo em três etapas. Primeiro, detecta padrões de gasto atípicos com três métodos estatísticos independentes e robustos (mediana/MAD, HHI de concentração de fornecedor), reduzindo 601 deputados a 253 sinalizados por 2+ métodos e 26 pelos 3 simultaneamente. Segundo, usa um LLM (Claude, via API Anthropic) com saída estruturada via Pydantic para classificar severidade e explicar cada caso em linguagem cidadã, sem nunca julgar reputação de pessoas. Terceiro, avalia o LLM em múltiplas camadas: consistência contra uma rubrica determinística e concordância contra rotulagem humana genuína (kappa de Cohen = 0,231; acurácia 40%). O achado central da avaliação é um viés de colapso do LLM: 70% dos casos foram classificados como severidade "média" e 84% como categoria "combinação", quase independentemente do sinal real — não ruído aleatório, mas uma tendência sistemática a respostas "seguras". O custo médio de classificação por caso é de US\$ 0,00525 (modelo claude-sonnet-4-6), o que torna barato escalar a classificação a milhares de casos. Todo número citado neste resumo também é gerado por um chunk de código executável no corpo deste documento.

Palavras-chave: Cota Parlamentar. Detecção de Anomalias. Modelos de Linguagem. Saída Estruturada. CRISP-DM.

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1 INTRODUCTION

The Chamber of Deputies publishes CEAP (Parliamentary Activity Exercise Allowance, *Cota para o Exercício da Atividade Parlamentar*) spending daily. Within the scope of this project (Jan/2025 to Jul/2026), that already amounts to 284,845 individual expense records and R\$345,090,001.79 — a volume that is humanly unauditably one entry at a time. Without a way to prioritize, any attempt at manual oversight is unfeasible: citizens, journalists, and oversight bodies have no way to review hundreds of thousands of rows to decide where it is worth looking more closely.

This work’s target audience comprises three groups: **citizens** interested in tracking how their representatives use the allowance; **journalists** who need concrete starting points for accountability reporting; and **auditors/oversight bodies** who already have an oversight process in place but benefit from prior triage that points to where to concentrate effort.

The goal is to **prioritize, not accuse**. The pipeline does not determine whether an expense is irregular — that requires human investigation, context, and the right to respond. What this work delivers is an objective statistical ranking of cases that deviate from the expected pattern, complemented by a classification and explanation layer in language accessible to the average citizen via an LLM, always with structured, validated output. The ultimate goal is to allocate the limited attention of those doing oversight more efficiently.

2 LITERATURE REVIEW

The direct precedent for this work, in the same domain (CEAP spending), is **Operação Serenata de Amor** (MUSSKOPF; CUDUCOS, 2016) (loosely, “Operation Love Serenade”): an open-source project started in 2016 by Irio Musskopf and Eduardo Cuducos that uses a machine-learning robot (“Rosie”) to compute a probability of irregularity in Parliamentary Allowance reimbursements and publish the findings. This work differs in three respects. First, it uses robust statistical detection (median/MAD) as an auditable layer prior to any classification, rather than a single probability score. Second, it uses an LLM only to classify and explain, in terms any reader can understand, cases already flagged statistically — never to decide on its own what is atypical. Third, it includes triangulation across independent methods and an evaluation of human-LLM agreement (Cohen’s kappa coefficient) as an explicit validation layer, absent from the original Rosie.

This work’s statistical layer follows **Iglewicz and Hoaglin (1993)**: median and median absolute deviation (MAD) as robust measures of central tendency and dispersion, and the “modified z-score” (factor 0.6745) as a practical rule for flagging outliers — a deliberate choice, since it is less sensitive to long tails than the classic mean/standard deviation, a problem well documented in parliamentary spending data itself.

To measure spending concentration in a single supplier, we use the **Herfindahl-Hirschman Index (HHI)** (HIRSCHMAN, 1964), originally a measure of market concentration in industrial economics, repurposed here to measure spending concentration per legislator.

To evaluate agreement between the LLM’s classification and human labeling, we use the **Cohen’s kappa coefficient** (COHEN, 1960), interpreted using the reference scale of **Landis and Koch (1977)**.

The LLM classification step follows the official Anthropic API documentation for output validated by a Pydantic schema (ANTHROPIC, 2026).

3 METHODOLOGY AND DATA

3.1 Sources and volume

Data come from the Chamber’s public annual files (<https://www.camara.leg.br/cotas/Ano-{ano}>) years 2025 (complete) and 2026 (partial, through July). Nulls concentrate in legitimately optional columns (txtPassageiro/txtTrecho, populated only for air travel) and in two peculiarities explicitly handled during data preparation: 7,191 records with negative vlrLiquido (genuine reversals, kept in the sum) and 45,635 records with no identifiable supplier (generic or missing CNPJ, excluded only from the supplier-concentration calculation).

3.2 Estimated token cost

The LLM classification step has a measured cost, not one estimated by analogy: using Anthropic’s (free) token-counting API on the prompts actually sent and the responses received for the 50 already-classified cases, the total cost was US\$0.2623, or **US\$ 0.00525 per case** (model claude-sonnet-4-6, US\$ 3.0 per million input tokens and US\$ 15.0 per million output tokens). The linear projection for the 253 prioritized cases (2+ converging methods) is approximately US\$ 1.33 — making pipeline extension financially trivial.

3.3 Mapping to CRISP-DM

CRISP-DM Phase	Deliverable in this project
1. Business Understanding	Problem, target audience, and success criteria (reports/business_understanding.md)
2. Data Understanding	Volume, types, nulls, token cost (reports/data_understanding.md, above)
3. Data Preparation	Cleaning, typing, flags with no row exclusion (reports/data_preparation.md)
4. Modeling	3 statistical detectors + deterministic baseline + LLM with fixed schema (reports/modeling.md)
5. Evaluation	Sensitivity, triangulation, LLM vs. rubric, LLM vs. human (reports/evaluation.md, reports/evaluation_llm.md)
6. Deployment	Prioritized list, citizen report, this paper (reports/relatorio_final.md)

3.4 LLM structured output

LLM classification uses a fixed Pydantic schema (`TriageAnomalia`), which forces the fields `severidade` (low/medium/high), `categoria_anomalia`, `explicacao_cidada` (max. 280 characters, no accusation), and `perguntas_auditoria` (2 to 3 verifiable audit questions) — see the full code in the Implementation section. The LLM only receives cases **already** statistically flagged; it never decides on its own what is atypical.

3.5 Methodological choices and trade-offs

The decisions below are neither neutral nor inevitable — they are worth making explicit, because they shape what the pipeline prioritizes and what it leaves out.

Domain: I chose CEAP over health spending (DataSUS/FNS), another public-spending domain of comparable volume that I mention in Next Steps as an extension. I chose CEAP because the Chamber’s annual files already come in a single, stable format, whereas health data would require integrating multiple sources before any analysis. Given the final project’s deadline, I prioritized delivering a complete pipeline (raw data → paper) over deepening the exploration of a domain that is more complex to integrate.

Detection method: I chose median + MAD (IGLEWICZ; HOAGLIN, 1993) over a machine-learning model (e.g., isolation forest) for the three detectors in `src/model.py`. In a domain where the result links public figures’ names to an attention ranking, a method whose decision cannot be explained in one sentence is not acceptable, even if it performed better on some anomaly-detection metric. Median and MAD are verifiable by hand by anyone with a calculator.

Division of labor between statistics and the LLM: “the LLM never accuses” is a design decision, not a workaround-able technical limitation. The LLM does not decide what is atypical: that has already been established statistically before any prompt. The `src/llm_triage.py` *system prompt* explicitly forbids any insinuation of wrongdoing, because the cost of a false positive here is not just any classification error: it is a person’s name attached to an accusation they never committed.

Error found during exploration: the first implementation of the generic-CNPJ filter compared against 12-digit codes instead of 14, returning zero records when in fact there are 8,527 (see `reports/data_understanding.md`). The fix only happened because a zero result was too suspicious to accept without checking.

Two-pass human labeling: the first labeling pass over the 50 evaluation cases used only the `category×peers` signal, a stricter criterion than the rubric’s definition (`reports/criterio_rotulagem.md`). The second pass reviewed the 2 cases where this changed the outcome. One of them, Bruno Ganem, remains a manually decided borderline case (severity “medium”, not “high”): concentration in a Facebook-ads supplier is a common pattern for parliamentary-outreach campaigns, a judgment call the rubric alone does not capture.

4 IMPLEMENTATION

4.1 Environment setup

This section walks, in code, through the same steps executed by the real scripts in `src/`: reading credentials, preparing the data, the three statistical detectors, and the call to the LLM with structured output. The first step is reading the API key from the environment (`.env`) — never written directly into the source code, per the project’s secrets rule.

```
# Imports and reading of environment keys (.env). NEVER write the key here.
import os
```

```

from pathlib import Path

import pandas as pd
from anthropic import Anthropic
from dotenv import load_dotenv
from pydantic import BaseModel, Field

load_dotenv(Path(__file__).resolve().parent / ".env")
client = Anthropic() # reads ANTHROPIC_API_KEY from the environment

```

4.2 Data collection and preparation

The real pipeline (`src/explore.py` → `src/prepare.py`) downloads, cleans, and types the 284,845 raw records without dropping a single row. The chunk below loads the already processed and verified result (`despesas_limpas.parquet`) to confirm the counts cited in Methodology:

```

print(f"Rows: {n_despesas:,} | Columns: {n_colunas_raw}")
print(f"Total spent (vlrLiquido): R$ {total_gasto:,.2f}")
print(f"Reversals (vlrLiquido < 0): {n_estornos:,}")
print(f"No identified supplier: {n_sem_fornecedor:,}")

```

```

Rows: 284,845 | Columns: 37
Total spent (vlrLiquido): R$ 345,090,001.79
Reversals (vlrLiquido < 0): 7,191
No identified supplier: 45,635

```

4.3 Statistical modeling (baseline)

Three independent statistical detectors (`src/model.py`), all based on median + MAD (robust score, threshold 3.5) or HHI (threshold 0.25). They serve as a **non-LLM baseline** for comparison: a deterministic rubric (`src/rubric_prelabel.py`) applies the same thresholds with no subjectivity, and is used in the evaluation as a consistency reference point for the LLM.

```

def robust_z(x: pd.Series) -> pd.Series:
    mediana = x.median()
    mad = (x - mediana).abs().median()
    escala = mad if mad > 0 else 1e-6 * (abs(mediana) + 1)
    return 0.6745 * (x - mediana) / escala

print(f"Legislators evaluated (>=1 detector): {n_deputados_total}")

```

```
print(f"Flagged by 2+ methods (high confidence): {n_alta_confianca}")
print(f"Flagged by all 3 methods simultaneously: {n_tres_metodos}")
```

Legislators evaluated (≥ 1 detector): 601
 Flagged by 2+ methods (high confidence): 253
 Flagged by all 3 methods simultaneously: 26

4.4 LLM modeling (structured output)

Pydantic schema and the real API call used in production (`src/llm_triage.py`) — shown as reference code (`eval: false`, already executed previously, with its real results loaded in the next chunk so as not to repeat a paid call). The code below is reproduced verbatim, including its Portuguese-language docstrings and field descriptions, since it is the actual production system operating for Brazilian citizens, journalists, and auditors:

```
from typing import Literal

class TriageAnomalia(BaseModel):
    severidade: Literal["baixa", "media", "alta"] = Field(
        description="Urgência de verificação humana, dada a magnitude e convergência dos sinais"
    )
    categoria_anomalia: Literal[
        "concentracao_fornecedor", "pico_temporal", "gasto_acima_pares", "combinacao"
    ] = Field(description="Tipo dominante do padrão detectado")
    explicacao_cidada: str = Field(max_length=280, description="Linguagem simples, SEM acusação")
    perguntas_auditoria: list[str] = Field(min_length=2, max_length=3)

MAX_RETRIES = 2 # 1 tentativa + 1 retry com feedback do erro de validação

def classificar(client, caso: str) -> tuple[TriageAnomalia | None, str]:
    """Chama a API e valida com Pydantic; retry com feedback em caso de erro."""
    feedback = ""
    for tentativa in range(1, MAX_RETRIES + 1):
        resposta = client.messages.create(
            model="claude-sonnet-4-6",
            max_tokens=600,
            system=SYSTEM_PROMPT,
            messages=[{"role": "user", "content": USER_PROMPT_TEMPLATE.format(caso=caso) + feedback}
        )
        try:
            dados = json.loads(resposta.content[0].text)
            return TriageAnomalia.model_validate(dados), f"ok_tentativa_{tentativa}"
        except Exception as e:
```

```
feedback = f"\n\nSua resposta anterior falhou na validação: {e}. Corrija."
return None, "nao_classificado" # falha final: registro marcado, não descartado
```

Real, already-classified result (reports/triage_llm.csv, loaded next — no API call is repeated in this render). The category and severity value counts below are the literal labels recorded by the real pipeline (in Portuguese), preserved as-is for traceability to the underlying data:

```
print(triage["status_llm"].value_counts().to_string())
print()
print(triage["severidade"].value_counts().to_string())
```

```
status_llm
ok_tentativa_1    50

severidade
media    35
baixa    10
alta      5
```

All 50 cases were classified successfully on the first attempt (ok_tentativa_1) — no Pydantic validation retry was necessary.

5 RESULTS

5.1 Prioritized list

Table 1 shows the highest-confidence cases: all three statistical methods converging. The full list reduces the universe from 601 legislators to 253 flagged by 2 or more methods — the central “business finding” of the Modeling/Evaluation phase.

5.2 Statistical evaluation: threshold sensitivity

The count decreases monotonically and gradually across all detectors (Table 2) — there is no abrupt jump near the default threshold, indicating that the choice of 3.5/0.25 is not an artificially sensitive point.

Table 1 – Top 10 cases prioritized by number of converging methods.

Legislator	Methods (0–3)	Dominant reason
Aguinaldo Ribeiro	3	gasto muito acima dos pares em 'PASSAGEM AÉREA - REEMBOLSO' (06/2025, ...
Benedita da Silva	3	gasto muito acima dos pares em 'HOSPEDAGEM ,EXCETO DO PARLAMENTAR NO D...
Bruno Ganem	3	gasto muito acima dos pares em 'ASSINATURA DE PUBLICAÇÕES' (08/2025, R...
Célio Silveira	3	gasto muito acima dos pares em 'DIVULGAÇÃO DA ATIVIDADE PARLAMENTAR.' ...
Elcione Barbalho	3	gasto muito acima dos pares em 'DIVULGAÇÃO DA ATIVIDADE PARLAMENTAR.' ...
Ely Santos	3	gasto muito acima dos pares em 'DIVULGAÇÃO DA ATIVIDADE PARLAMENTAR.' ...
Erika Kokay	3	gasto muito acima dos pares em 'SERVIÇO DE TÁXI, PEDÁGIO E ESTACIONAME...
Fabio Schiochet	3	gasto muito acima dos pares em 'HOSPEDAGEM ,EXCETO DO PARLAMENTAR NO D...
Gabriel Mota	3	gasto muito acima dos pares em 'MANUTENÇÃO DE ESCRITÓRIO DE APOIO À AT...
Gilson Marques	3	gasto muito acima dos pares em 'HOSPEDAGEM ,EXCETO DO PARLAMENTAR NO D...

Source: own elaboration from Chamber of Deputies open data (CEAP, 2025-2026).

Data via `reports/lista_priorizada.csv` (`src/deploy.py`). Full list with 253 cases in the original file. Names and reason text are reproduced in the original Portuguese, as generated by the real pipeline.

Table 2 – Sensitivity of flag counts to alternative thresholds (recomputed live).

Detector	Low threshold	Default	High threshold
Category×peers (score)	1,785	1,296	1,004
Temporal peak (score)	770	582	457
Supplier concentration (HHI)	326	141	71

Source: own elaboration from Chamber of Deputies open data (CEAP, 2025-2026).

Thresholds: score 3.0/3.5/4.0; HHI 0.15/0.25/0.35. Recomputed live from the CSVs in `data/processed/`.

Table 3 – Confusion matrix: severity (human) vs. severity (LLM), n=50.

Human \ LLM	High	Medium	Low
High	5	20	0
Medium	0	6	1
Low	0	9	9

Source: own elaboration from Chamber of Deputies open data (CEAP, 2025-2026). Cross-tabulation of reports/rotulos_humanos.csv and reports/triage_llm.csv (src/evaluate_llm.py), recomputed live.

5.3 LLM evaluation: rubric vs. human

Central finding of this evaluation: the LLM collapses toward the most comfortable label instead of reflecting the actual signal. Compared against genuine human labeling (n=50), the LLM achieves only 40% severity accuracy and **Cohen’s kappa = 0.231** (Table 3) — on the Landis and Koch (1977) scale, “fair” agreement, not “substantial”. Category agreement is even lower: 28%.

The cause is not random noise: the LLM classified severity as “**medium**” in 70% of cases and category as “**combination**” in 84%, almost independently of the reference value (human or rubric). Against the deterministic rubric, accuracy drops to 24% — even lower, because the rubric never “hesitates”: it is 100% deterministic, and the LLM, without knowing its thresholds, converges toward the seemingly safe middle ground.

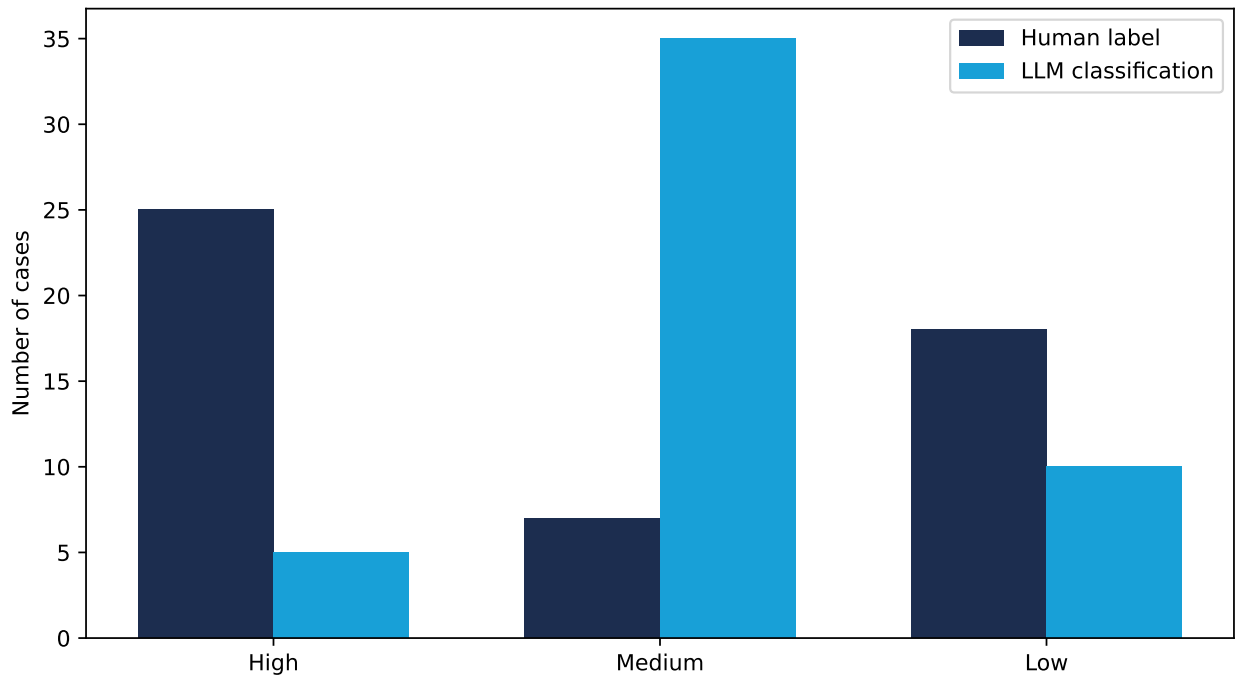


Figure 1 – Severity distribution: human label vs. LLM classification (n=50). Source: own elaboration from Chamber of Deputies open data (CEAP, 2025-2026).

Figure Figure 1 makes the collapse clear: human labeling is distributed more evenly across the three classes, while the LLM concentrates the overwhelming majority on “medium” regardless of the case. **Improvement proposed for a future iteration:** inject the rubric’s numeric thresholds directly into the LLM prompt (R\$50,000 / R\$10,000 / HHI 0.5 / 5+ flags / partial month), so that classification stops being a blind qualitative judgment and instead anchors on the same numbers that already guide the deterministic rubric — without transferring to the LLM the decision about what is atypical, only giving it the explicit criteria to decide with more rigor.

5.4 Cost per case and update plan

The cost of US\$0.00525 per case (measured via `count_tokens`, not estimated by analogy) is negligible at scale. **Update plan:** the Chamber updates the data daily, but there is no need to reprocess at that frequency — a monthly or quarterly cadence is sufficient. When reprocessing, review whether the thresholds in `src/model.py` (`SCORE_THRESHOLD = 3.5`, `HHI_THRESHOLD = 0.25`) remain adequate in light of the sensitivity analysis (Table 2), since flag volume may grow as more 2026 months arrive.

6 NEXT STEPS

- **Inject the rubric into the LLM prompt:** give the model explicit numeric thresholds (R\$50,000, R\$10,000, HHI 0.5, 5+ flags, partial month) to reduce the collapse bias identified in the evaluation, without delegating to the LLM the decision about what is atypical.
- **Extension to health spending (DataSUS/FNS):** the same statistical pipeline (median/MAD, HHI, temporal peaks) applies directly to National Health Fund transfers by municipality, another auditable, high-volume public-spending domain.
- **Monthly report automation:** schedule the full pipeline run (explore → prepare → model → evaluate → deploy → llm_triage) followed by rendering this paper, delivering an updated version with no manual intervention.

7 CONCLUSION

This work delivered a complete, reproducible CRISP-DM pipeline to prioritize anomalous Parliamentary Allowance expenses: three robust statistical detectors reduce 601 legislators to 253 priority-attention cases; an LLM with output validated by a Pydantic schema classifies and explains each one in plain language, free of technical jargon, at a cost of US\$ 0.00525 per case; and a multi-layer evaluation (sensitivity, triangulation, deterministic rubric, human labeling) shows that the LLM still leans toward the more cautious judgment ($\kappa = 0.231$, “fair” rather than “substantial” on the Landis and Koch (1977) scale) instead of reflecting each case’s actual signal.

The pipeline’s statistical part (three detectors + triangulation) is reproducible and auditable: the reduction to 253 priority cases, described above, already eliminates the need to manually review hundreds of thousands of rows. LLM classification, on the other hand, should not be used today without human review of the `explicacao_cidada` it generates: a κ of 0.231 is too low to treat the LLM’s output as a verdict. The next iteration described in Next Steps — injecting the rubric’s thresholds into the prompt — targets this specific cause; until it is tested, this work’s value lies in the statistical layer, not the language layer.

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